Derive $\mathcal{H}$

$\mathcal{L}(q_i, \dot{q}_i, t)$ ← explicit

$$\frac{d\mathcal{L}}{dt} = \left(\frac{\partial \mathcal{L}}{\partial q_i}\dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\ddot{q}_i\right) + \cancel{\frac{d\mathcal{L}}{dt}}^{\,0 \text{ (if implicit)}}$$

$$= \dot{p}_i \dot{q}_i + p_i \ddot{q}_i$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt}(p_i \dot{q}_i)$$

$$\frac{d}{dt}(\mathcal{L} - p_i \dot{q}_i) = 0$$

$$\boxed{\mathcal{L} - p_i \dot{q}_i = \text{const}} = \mathcal{H}$$

$$\frac{1}{2}m\dot{q}_i^2 - V(q_i) - m\dot{q}_i^2 - \left(\frac{1}{2}m\dot{q}_i^2 + V(q_i)\right) = \text{const.}$$

also. E_g = const

$$\mathcal{H} = \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

Example - 1.1

$\rightarrow \quad \dot{\xi} = \sqrt{km}\,\dot{x}$

$\Rightarrow \quad \dot{\xi}^2 = km\dot{x}^2$

$\Rightarrow \quad \dot{x}^2 = \frac{\dot{\xi}^2}{km}$

$\omega = \sqrt{\frac{k}{m}}$

$$\mathcal{L} = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2}$$

$$= \frac{\dot{\xi}^2}{2\omega} - \frac{\omega}{2}\xi^2$$

$$\boxed{\mathcal{L} = \frac{\dot{\xi}^2}{2\omega} - \frac{\omega\xi^2}{2}}$$

Example - 1.2 ✓ convert to $\mathcal{H}$

$$\mathcal{L} = \frac{\dot{\xi}^2}{2\omega} - \frac{\omega}{2}\xi^2$$

$$\mathcal{H} = P\dot{\xi} - \mathcal{L}$$

$$= P\dot{\xi} - \frac{\dot{\xi}^2}{2\omega} + \frac{\omega}{2}\xi^2$$

$$\mathcal{H} = \frac{\dot{\xi}^2}{\omega} - \frac{\dot{\xi}^2}{2\omega} + \frac{\omega}{2}\xi^2$$

$$= \omega\frac{\dot{\xi}^2}{2\omega^2} + \frac{\omega}{2}\xi^2$$

$$\mathcal{H} = \frac{P^2\omega}{2} + \frac{\xi^2\omega}{2} \quad \left[\because P = \frac{\dot{\xi}}{\omega}\right]$$

$$\mathcal{H} = \frac{\omega}{2}(P^2 + \xi^2)$$

Derive Hamilton's Eq.^{n}

$$\mathcal{H} = \sum_i (p_i \dot{q}_i) - \mathcal{L}(q_i, \dot{q}_i)$$

$$\delta\mathcal{H} = \sum_i \left(\dot{q}_i \delta p_i + \cancel{p_i \delta \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i - \cancel{\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i}\right)$$

$$\delta\mathcal{H} = \sum_i \left(\dot{q}_i \delta p_i - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i\right) \quad \frac{\partial \mathcal{L}}{\partial q_i} = \dot{p}_i \quad \text{— (i)}$$

$$\delta\mathcal{H}(q_i, p_i) = \sum_i \left(\frac{\partial \mathcal{H}}{\partial p_i}\delta p_i + \frac{\partial \mathcal{H}}{\partial q_i}\delta q_i\right) \quad \text{— (ii)}$$

Comparing (i) & (ii) :

$$\boxed{\frac{\partial \mathcal{H}}{\partial p_i} = \dot{q}_i \quad ; \quad \frac{\partial \mathcal{H}}{\partial q_i} = -\dot{p}_i}$$

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